



Research



Nonlinear dynamics response of porous functionally graded annular plates using modified higher order shear deformation theory

Hojat Jafary¹ · Mohsen Taghizadeh¹

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Abstract

In this investigation, the nonlinear dynamics of porous annular plates that have been functionally graded are examined under varying time-dependent loads. Both simple supported and fully clamped boundary conditions are taken into account. The mechanical properties of the functionally graded plate in its thickness are considered according to the distribution function of the modified law. In addition, porosity, as a stress-relieving property, is also applied throughout the plate thickness by different functions and pore volume fractions. By adopting Hamilton's principle, the equations of motion are obtained based on the modified higher-order shear deformation plate theory. Then these partial differential equations are solved using the viscous dynamic relaxation method in conjunction with Newmark's implicit integration method. The present findings are compared and successfully confirmed with those available in the literature. Finally, the effects of some key factors such as porosity distribution type, pore volume fraction, power-law index, loading conditions, and thickness-to-radius ratio on the dynamic behavior of both simple and fully clamped plates have been studied in detail. The study's findings indicate that porosity pattern X, characterized by a greater prevalence of interconnected pores, exhibited enhanced resistance to deflection. Conversely, pattern O, featuring a more uniformly distributed pore size, effectively mitigated stress in FG porous materials. These observations offer valuable insights for optimizing plate design and elevating its performance.

Keywords Annular plates · Functionally graded porous materials · Modified higher-order shear deformation plate theory · Viscous dynamic relaxation method · Implicit Newmark integration procedure

1 Introduction

In the field of materials science, functionally graded (FG) materials are distinguished by their gradual transition in composition and structure. Various materials are designed based on the specific application they will be used for, and can be manufactured using particle, continuous, or layered processing methods. The functionally graded materials (FGMs) are made up of components with different properties that work together to provide excellent corrosion resistance, flexibility, and formability. This study provides valuable insights for the design and optimization

of porous functionally graded annular plates, which have potential applications in aerospace structures, biomedical implants, and civil infrastructures.

Porosity represents desirable features such as permeability, creep resistance, dielectric strength, and fragility reduction that have received many researchers' attention during the past years. Zhang and Zhou studied a nonlinear circular FG plate using higher-order shear deformation plate theory (HSDT) and solved the governing equations of equilibrium through the multi-term scribe method [1]. The influences of edge support, thermal conditions, and volume fraction index are examined in this paper.

✉ Mohsen Taghizadeh, m.taghizadeh@hsu.ac.ir | ¹Department of Mechanical Engineering, Hakim Sabzevari University, Sabzevar, Iran.



Shojaeefard et al. [2] employed the generalized differential quadrature method (GDQM) to numerically study the linear free vibrations and buckling behavior of temperature-dependent FG porous (FGP) circular micro-plate based on the classical (CPT) as well as the first-order shear deformation plate theories (FSDT). They accounted for the influences of small scales, power-law index and temperature change. Zhao et al. [3] employed the linear FSDT together with a direct variational method to perform dynamic analysis of the behavior of annular, circular, and sector plates made of FG porous material, taking into account general elastic boundary conditions. Lu et al. [4] studied FG annular and circular plates based on a combination of FSDT, the virtual spring technique, the multi-segment partition method, and Rayleigh–Ritz semi-analysis with general conditions. Babaei et al. [5] investigated the free vibrations of thick annular functionally graded porous (FGP) sector plates and cylindrical shells, obtaining their natural frequencies and mode-shapes. The study investigated how natural frequencies and transient response are impacted by different boundary conditions, porosity coefficients, thickness-to-radius ratio, sector angle, and additional scattering of porosity. Arshid et al. [6] analyzed the behavior of porous nanocomposite annular sector plates with FG graphene platelets (GPL) reinforcements using both static and dynamic analysis. They employed the modified strain gradient theory and the FSDT to derive the equations of motion, which were then solved using the GDQM. The study investigated the effects of various parameters, including GPL fraction, porosity distribution patterns, size, and GPL geometric parameters. The study results provide insights into the behavior of these types of plates under different conditions. Rad [7] conducted a study on the mechanics of non-uniform two-dimensional flat geometrically-perfect (FGP) circular plates, considering gradient elastic foundations with frictional forces. The 3D pore elasticity theory was used to derive governing equations of motion which were solved using a state space-based differential quadrature method (DQM). The results indicated that the ascetic, torsional couples and rotational interaction of the elastic foundation play an important role in the plate behavior. Asemi et al. [8] examined the static, dynamic, and free vibration behaviors of FGP annular sector plates reinforced with graphene nanoplatelets (GNPs). Hamilton's principle together with the FSDT were implemented in order to derive the finite element governing equations of motion. This study considered the impacts of porosity, GPL distribution and weight fraction, sector angles, and boundary conditions. Arshid and Khorshidvand [9] studied the free vibrations of a FGP circular plate with an integrated piezoelectric actuator. They assumed the plate to be thin and neglected the influence of shear deformations. The CPT and Hamilton's principle were

utilized to derive the equations of motion, which were then solved using the DQM. The study examined the impact of several important factors, including thickness ratio, porosity, piezoelectric actuation, and pore distributions and accumulations, on the natural frequencies, radial stresses, and hoop stresses. Jankowski and Žur [10] analyzed free vibrations of FGP circular plates using CPT. Over the thickness of the plate, there were assumed two uniform and non-uniform porosity patterns. Investigating the influences of porosity, power-law index and various boundary conditions on the dimensionless frequencies of the circular plate was the objective of this research. Vinyas [11] studied the frequency response of FGP electromagnetic elastic circular and annular plates using HSDT. Aside from investigating the influences of some key parameters such as the gradient index and porosity volume fraction, this paper discussed the effect of various plate electromagnetic boundary conditions on natural frequencies for the first time. Babaei et al. [12] studied the dynamical behavior of a rotating thick truncated cone made of FGP. Three different porosity patterns along the thickness of the cone were assumed in this work. This paper utilizes a graded finite element method along with the Newmark integration technique. They observed that the natural frequencies increase as the Skempton coefficient increases. Ghugal and Shimpi [13] investigated the buckling behavior of moderately thick porous annular sector plates made of FGM. The governing equations were derived using Sanders's nonlinear strain–displacement equations and were decoupled using the four auxiliary functions method. Key parameters considered in the study included the thickness-to-radius ratio, plate porosity coefficient, and boundary conditions. A study conducted by Arshid et al. [14] investigated the phenomenon of thermal buckling in nanoplates reinforced with graphene and porous nanocomposites. The study focused on the behavior of these materials under thermal loading conditions and analyzed their performance in terms of thermal buckling. Micromechanical models were employed to determine the effective properties of the structure, and the governing equations were solved using the generalized differential quadrature method. The results indicate that an increase in the weight fraction of GNPs by 1% can cause a reduction in the critical buckling temperature by 7–15%. On the other hand, an increase in the porosity coefficient by 0.7 can improve the thermal buckling behavior by 20–50%. This study provides a benchmark for designing engineering structures with enhanced properties, particularly in aerospace engineering where metal foams are frequently utilized. Jabbari et al. [15] explored the thermal buckling analysis of thin circular plates composed of saturated porous material with a functionally graded structure. The plate had material properties that changed through its thickness and was secured

with clamped edges. Geometrical nonlinearities were accounted for by applying the Love–Kirchhoff hypothesis. Equilibrium and stability equations were then derived via a variational approach to ascertain prebuckling and critical buckling temperatures. These equations were formulated based on the Sanders non-linear strain–displacement relation. The porous nature of the plate, characterized by fluid-saturated pores, was considered, and the study delved into how pore distribution and thermal factors influenced the critical buckling temperature. Khatounabadi and Jafary [16] undertook the inaugural analysis of low-velocity impacts on circular plates with FGP reinforcement employing a gradient porosity law and the FSDT. To simulate the contact force between the impactor and the plate, they employed a customized Hertzian contact law. Solving the governing equations of motion involved the use of the finite element method in combination with the Picard method. Their findings illustrated the significance of the porosity pattern on the temporal evolution of central deflections and the contact force experienced by the plate, respectively. Zhong et al. [17] proposed a semi-analytical model for predicting the vibration behavior of annular porous plates. To theoretically establish a model for an elastically restrained annular plate, they combined a multi-segment technique integrated into a variational approach. The effects of material and geometry parameters on the vibration characteristics of porous plates are discussed in this paper. The investigation into the natural vibration characteristics of circular and annular plates with functionally graded porosity and bidirectional circular and annular plates with FG materials was carried out by Vasara et al. [18, 19]. Utilizing the FSDT in conjunction with Hamilton's principle, they formulated and solved the equations of motion employing the differential quadrature method. They investigated the natural frequencies of circular and annular plates with functionally graded porosity, considering factors such as the thickness-to-radius ratio, porosity distribution, and boundary conditions. The outcomes highlighted the substantial influence of these parameters on the mechanical behavior of porous bidirectional functionally graded plates. Babaei et al. [20] studied torsional buckling in thick truncated conical shell panels supported on a Winkler-type elastic medium. The shells used in this study are made of porous metal reinforced with graphene platelets (GPLs) with varying porosity and nanofiller content. They explored different porosity distribution functions and GPL patterns. Material properties are estimated using advanced models. The analysis employs numerical methods to compute torsional buckling forces, considering various parameters. The primary goal is to optimize porosity distribution and GPL patterns and assess the impact of nanofillers on buckling forces. Results show that the GPL-X pattern with porosity distribution 1 yields the

highest buckling loads, with nanofillers increasing loads by up to 100%. This research advances structural performance through tailored materials and nanofiller integration. Khatoonabadi et al. [21] investigated on a novel analysis of shear buckling in functionally graded porous annular sector plates reinforced with GPLs. The plates consist of a layered model with varying graphene nanoplatelet dispersion in a metallic matrix, incorporating open-cell interior pores. Effective mechanical properties are determined using the extended rule of mixture and modified Halpin–Tsai models. Governing equations are derived based on 3D elasticity theory and the generalized geometric stiffness concept, and finite element analysis is employed to solve these equations. They explored the impact of various parameters, including porosity distribution, porosity coefficient, GPL dispersion patterns, GPL weight fraction, boundary conditions, and sector angles on shear buckling loads in these plates and also represented the first investigation of its kind and offers valuable insights into the behavior of these nanocomposite structures. Babaei et al. [22] reviewed the mechanical properties of functionally graded saturated porous structures, providing a comprehensive analysis of their mechanical behavior in saturated conditions, and discussed the challenges and potential avenues for future research in the field of functionally graded porous structures. Chen et al. [23] explored the vibration and damping characteristics of a deep spherical sandwich shell made with electrorheological (ER) fluid in the core and graphene-enhanced nanocomposite skins. Their study combines deep shell theory and linear constitutive laws to formulate governing motion equations. These equations are solved using a semi-analytical approach involving trigonometric expansion (TE) and the generalized differential quadrature (GDQ) method. The analysis covers the first ten vibrational angular modes, considering various parameters to investigate frequencies and loss factors. Noroozi and Malekzadeh [24] examined the nonlinear response of multilayer graphene-reinforced composite skew plates under moving loads. They employed advanced mathematical techniques and numerical methods for analysis, showing rapid convergence and high accuracy. The research explores the influence of different factors on the nonlinear response, indicating that the nonlinear theory results in a lower dynamic magnification factor compared to the linear theory. Nguyen et al. [25] studied on an analytical approach to examine the nonlinear thermo-mechanical buckling of functionally graded graphene platelet reinforced composite structures. These structures are subjected to external pressure and thermal loads, and rest on a nonlinear elastic medium. The Ritz energy method is used to investigate the post buckling response and critical buckling load. They also explored the effects of varying medium stiffnesses, material

parameters, and geometrical dimensions on the behavior of structures. Wu et al. [26] analyzed the thermal post buckling behavior of FG graphene platelets-reinforced composite annular plates with geometric imperfections using FSDT and considering von Kármán geometric nonlinearity. They used an imperfection model, 3D Halpin–Tsai model, and nonlinear governing equations. Parametric studies reveal that geometric imperfections reduce thermal post buckling resistance, increasing as imperfection amplitude/half-wave number increases. Dogan [27] investigated the dynamic response of viscoelastic plates reinforced with FG carbon nanotube-reinforced composite material (FG-CNTRC) under dynamic loads. The research investigates single-walled CNTs configurations and their influence on the behavior of the plates. Moreover, the study demonstrates that the Laplace technique can be used to solve problems precisely, effectively, and easily, without the need for free vibration frequencies or modes. Sadraee Far et al. [28] examined the bending of a 3D FG carbon nanotube reinforced composite cylindrical shell under internal pressure. The shell was reinforced along the longitudinal axis and the distribution of CNTs is uniform. The governing equations are developed using the principle of virtual displacements and differential quadrature method. Finally, the results show that non-dimensional maximum deflections of FG distributions are higher than uniform distributions. You et al. [29] studied the transient coupled poro/thermoelectricity response of FG CNT reinforced (FG-CNTR) nanocomposite panels surrounded by a non-polynomial elastic foundation. The nanocomposite structure was subjected to initial mechanical loads and thermal shock loading. The governing equations are developed using 3D elasticity theory and Navier approach and the energy balance equation is used to determine the temperature gradient of the panel. The study reveals that increasing the volume-fraction of CNT negatively impacts transverse shear stress variation. Minh et al. [30] presented the results of the stability analysis of FG-CNTRC plates under combined loads, including external pressure and axial compression loads with uniformly distributed temperature rises. They used HSDT, von Kármán's geometrical nonlinearities, and anisotropic beam theory to develop a new smeared stiffener technique. They also discussed the influence of stiffeners, nonlinear elastic foundation, and geometrical features.

In various industries and applications, researchers have been focusing on combining porosity in structures to achieve the most efficient results. A number of analytical and modeling investigations have been carried out in order to assess the impact of porosity and its various characteristics, such as vibration analysis, frequency, buckling, and bending, on the stiffness and other material properties of structures. This paper

examines the dynamic response of FGP annular plates under dynamic loading, utilizing a novel modified high-order shear deformation theory (MHSDT). The study takes into account nonlinear Von-Karman relations with both clamped and simply supported boundary conditions. The FGP plate exhibits three distinct porosity patterns, namely one uniform dispersion and two parabolic non-uniform dispersions. The material properties of the plate are modeled using a modified power-law equation that varies throughout the thickness of the plate. To solve the equations of motion, the viscous dynamic relaxation method and Newmark integration technique are employed for time-discretized relations.

2 Theoretical formulation of the FGP material

Figure 1 illustrates an annular FG porous plate with inner radius r_i and outer radius of r_o , and h represents the total thickness of the porous plate in polar coordinates. The origin of the plate is assumed to be at the center of the plate.

Different types of porosity dispersions used in this study are shown in Fig. 2. An even and two uneven patterns known as V_p :

Even pattern

$$V_p = 1 \quad (1)$$

Uneven patterns:

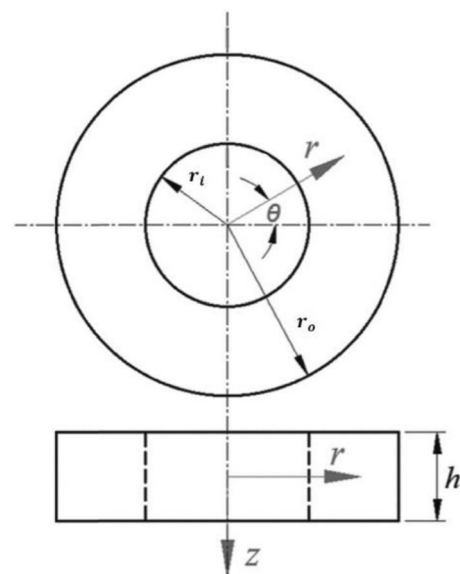


Fig. 1 Schematic of the annular FG porous plate

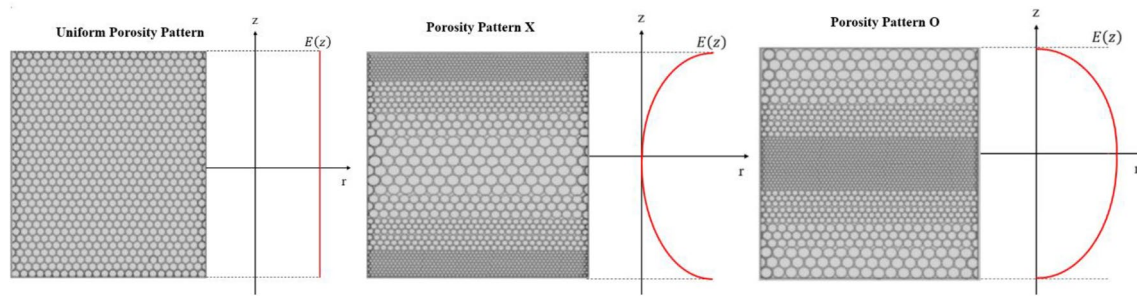


Fig. 2 Physical illustration of the FG porous annular plate with different porosity dispersions

Pattern X:

$$V_p = 1 - 1 \cos \left[\frac{|z|\pi}{h} \right] \quad (2)$$

Pattern O:

$$V_p = \cos \left[\frac{|z|\pi}{h} \right] \quad (3)$$

2.1 Functionally graded porous annular plate

Based on the modified rule of mixture, the constituent material properties of the porous FG plate can be formulated and proposed as [31]:

$$P_f = P_c \left(V_c - \frac{\zeta}{2} \right) + P_m \left(V_m - \frac{\zeta}{2} \right), \quad (4)$$

hereafter ζ , subscripts c and m indicate the porosity volume fraction, ceramic and metal, respectively. According to [31] $\zeta \ll 1$.

Also, V_c and V_m are ceramic and metal volume fractions. The association of volume fractions of metal and ceramic can be written as:

$$V_c + V_m = 1, \quad V_c = \left(\frac{z}{h} + \frac{1}{2} \right)^\alpha, \quad \alpha \geq 0 \quad (5)$$

where z is the distance from the middle surface and α denotes the power law index.

On the basis of a modified power-law model, modulus of elasticity and density of the plate for even porosity pattern can be defined as:

$$E(z) = (E_c - E_m) \left(\frac{z}{h} + \frac{1}{2} \right)^\alpha + E_m - \frac{\zeta}{2} (E_c - E_m), \quad (6)$$

$$\rho(z) = (\rho_c - \rho_m) \left(\frac{z}{h} + \frac{1}{2} \right)^\alpha + \rho_m - \frac{\zeta}{2} (\rho_c - \rho_m). \quad (7)$$

In the same way, $E(z)$ and $\rho(z)$ are defined for the uneven X pattern as:

$$E(z) = (E_c - E_m) \left(\frac{z}{h} + \frac{1}{2} \right)^\alpha + E_m - \frac{\zeta}{2} (E_c - E_m) \left(1 - 1 \cos \left[\frac{|z|\pi}{h} \right] \right), \quad (8)$$

$$\rho(z) = (\rho_c - \rho_m) \left(\frac{z}{h} + \frac{1}{2} \right)^\alpha + \rho_m - \frac{\zeta}{2} (\rho_c - \rho_m) \left(1 - 1 \cos \left[\frac{|z|\pi}{h} \right] \right), \quad (9)$$

and for the uneven O pattern as:

$$E(z) = (E_c - E_m) \left(\frac{z}{h} + \frac{1}{2} \right)^\alpha + E_m - \frac{\zeta}{2} (E_c - E_m) \left(\cos \left[\frac{|z|\pi}{h} \right] \right), \quad (10)$$

$$\rho(z) = (\rho_c - \rho_m) \left(\frac{z}{h} + \frac{1}{2} \right)^\alpha + \rho_m - \frac{\zeta}{2} (\rho_c - \rho_m) \left(\cos \left[\frac{|z|\pi}{h} \right] \right). \quad (11)$$

The continuously varying composition of the material is such that from the bottom surface ($z/h = -0.5$) to top surface ($z/h = 0.5$), the characterization of ceramic changes into metal. It is notable that the Poisson's ratio, ν is constant through the thickness.

The material properties of Aluminum and Zirconia of the plates used in this study are summarized in Table 1 [32].

3 Equations of motion

An outstanding and innovative feature of this work is the use of a modified high-order shear deformation theory for annular plates [33, 34]. In the displacement field of

Table 1 Material properties of ceramic and metal constituents of the FGP plates

Material	Elastic moduli (GPa)	Poisson's ratio	Density (Kg/m ³)
Ceramic (ZrO ₂)	200	0.3	7850
Metal (Al)	70	0.3	3960

shear theories of higher order than the first order, components appear that are physically difficult to interpret. Because of the axial symmetry, there are no changes in the θ direction [34]. The displacement field according to the modified-higher order shear deformation theory is defined as follows:

$$\begin{Bmatrix} u(r, z, t) \\ 0 \\ w(r, z, t) \end{Bmatrix} = \begin{Bmatrix} u_0(r, t) \\ 0 \\ w_0(r, t) \end{Bmatrix} + z \begin{Bmatrix} \varphi_r(r, t) \\ 0 \\ 0 \end{Bmatrix} + f(z) \begin{Bmatrix} \psi_r(r, t) \\ 0 \\ 0 \end{Bmatrix}. \quad (12)$$

Based on this displacement field, quadratic changes in shear strain along the thickness are considered, which are zero at the top and bottom of the plane. Since the shear strain changes in the thickness are very similar to their true state, therefore, there is no need for a shear correction factor. u and w are respectively the displacement components in the direction of the radius and thickness, r and z which are attached to each point of the plane at a distance of z from the middle distance of the plane. Also, u_0 and w_0 are the displacement components of the mid-plane based on the variables r , t and φ_r is the rotation of the component around the radial axis. ψ_r is a mathematical term that cannot be defined physically. One of the potential benefits of this theory is the simplicity of defining the degree of the theory using the $f(z)$ component, so that the desired function can be considered according to the demand of the problem. For example, considering z^3 as the displacement function, a third-order theory would be chosen for the concern. It should be stated that, in this study the z^3 is taken as the desired function for the $f(z)$ component.

The strain field in terms of displacement components in Eq. (12) according to the MHSMT is expressed as follows:

$$\begin{Bmatrix} \varepsilon_{rr} \\ \varepsilon_{\theta\theta} \\ \gamma_{rz} \end{Bmatrix} = \begin{Bmatrix} \frac{\partial u_0}{\partial r} + \frac{1}{2} \left(\frac{\partial w_0}{\partial r} \right)^2 + z \frac{\partial \varphi_r}{\partial r} + f(z) \frac{\partial \psi_r}{\partial r} \\ \frac{u_0}{r} + z \frac{\varphi_r}{r} + \frac{f(z)}{r} \psi_r \\ \frac{\partial w_0}{\partial r} + \varphi_r + f'(z) \psi_r \end{Bmatrix}, \quad -\frac{h}{2} < z < \frac{h}{2}, \quad (13)$$

where the perpendicular strain terms ε_{rr} and $\varepsilon_{\theta\theta}$ are in r and θ directions, respectively, and γ_{rz} is the shear strain. The relationship between stress and strain for an annular plate, based on the MHSMT whose matrix mode is mentioned in Hooke's law, is written as follows:

$$\begin{Bmatrix} \sigma_{rr} \\ \sigma_{\theta\theta} \\ \sigma_{rz} \end{Bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_{rr} \\ \varepsilon_{\theta\theta} \\ \gamma_{rz} \end{Bmatrix}, \quad (14)$$

where:

$$Q_{11} = \frac{E(z)}{(1-\nu^2)}, \quad Q_{22} = \frac{E(z)}{(1-\nu^2)}, \quad Q_{12} = \frac{E(z)\nu}{(1-\nu^2)}, \quad Q_{66} = \frac{E(z)}{2(1+\nu)}, \quad (15)$$

in which ν represents the poisson's ratio.

The resultant force, moment and other higher order resultant terms are defined as:

$$(N_{ij}, M_{ij}, R_{ij}) = \int_{-h/2}^{h/2} (1, z, f(z)) \sigma_{ij} dz, \quad ij = rr, \theta\theta, \quad (16)$$

$$(Q_{rz}, R_{rz}) = \int_{-h/2}^{h/2} (\sigma_{rz}, f'(z) \sigma_{rz}) dz. \quad (17)$$

The elastic coefficient matrices based on the MHSMT are expressed as follows:

$$(A_{mn}, B_{mn}, C_{mn}, F_{mn}, G_{mn}, H_{mn}) = \int_{-h/2}^{h/2} Q_{mn} (1, z, f(z), z^2, zf(z), f^2(z)) dz, \quad mn = 11, 12, 22, \quad (18)$$

$$(D, E, I) = \int_{-h/2}^{h/2} Q_{66} (1, f'(z), f'(z)^2) dz. \quad (19)$$

Substituting Eqs. (13) and (14) into (16) and (17) leads to the following resultants:

$$N_{rr} = A \left\{ \frac{\partial u_0}{\partial r} + \frac{1}{2} \left(\frac{\partial w_0}{\partial r} \right)^2 + \nu \frac{u_0}{r} \right\} + B \left\{ \frac{\partial \varphi_r}{\partial r} + \nu \frac{\varphi_r}{r} \right\} + C \left\{ \frac{\partial \psi_r}{\partial r} + \nu \frac{\psi_r}{r} \right\}, \quad (20)$$

$$N_{\theta\theta} = A \left\{ \nu \frac{\partial u_0}{\partial r} + \frac{\nu}{2} \left(\frac{\partial w_0}{\partial r} \right)^2 + \frac{u}{r} \right\} + B \left\{ \nu \frac{\partial \varphi_r}{\partial r} + \frac{\varphi_r}{r} \right\} + C \left\{ \nu \frac{\partial \psi_r}{\partial r} + \frac{\psi_r}{r} \right\}, \quad (21)$$

$$M_{rr} = B \left\{ \frac{\partial u_0}{\partial r} + \frac{1}{2} \left(\frac{\partial w_0}{\partial r} \right)^2 + \nu \frac{u_0}{r} \right\} + F \left\{ \frac{\partial \varphi_r}{\partial r} + \nu \frac{\varphi_r}{r} \right\} + G \left\{ \frac{\partial \psi_r}{\partial r} + \nu \frac{\psi_r}{r} \right\}, \quad (22)$$

$$M_{\theta\theta} = B \left\{ v \frac{\partial u_0}{\partial r} + \frac{v}{2} \left(\frac{\partial w_0}{\partial r} \right)^2 + \frac{u_0}{r} \right\} + F \left\{ v \frac{\partial \varphi_r}{\partial r} + \frac{\varphi_r}{r} \right\} + G \left\{ v \frac{\partial \psi_r}{\partial r} + \frac{\psi_r}{r} \right\}, \quad (23)$$

$$R_{rr} = C \left\{ \frac{\partial u_0}{\partial r} + \frac{1}{2} \left(\frac{\partial w_0}{\partial r} \right)^2 + v \frac{u_0}{r} \right\} + G \left\{ \frac{\partial \varphi_r}{\partial r} + v \frac{\varphi_r}{r} \right\} + H \left\{ \frac{\partial \psi_r}{\partial r} + v \frac{\psi_r}{r} \right\}, \quad (24)$$

$$R_{\theta\theta} = C \left\{ v \frac{\partial u_0}{\partial r} + \frac{v}{2} \left(\frac{\partial w_0}{\partial r} \right)^2 + \frac{u_0}{r} \right\} + G \left\{ v \frac{\partial \varphi_r}{\partial r} + \frac{\varphi_r}{r} \right\} + H \left\{ v \frac{\partial \psi_r}{\partial r} + \frac{\psi_r}{r} \right\}, \quad (25)$$

$$Q_{rz} = D \left\{ \varphi_r + \frac{\partial w_0}{\partial r} \right\} + E \{ \psi_r \}, \quad (26)$$

$$R_{rz} = E \left\{ \varphi_r + \frac{\partial w_0}{\partial r} \right\} + I \{ \psi_r \}. \quad (27)$$

4 Hamilton's Principle

The principle of virtual displacement (virtual work) states that; If a continuous body is in equilibrium, the virtual work of all real forces due to the virtual displacement will be zero. Hamilton's principle is a generalization of this principle for dynamic systems. According to Hamilton's principle, the analyzed system is defined using two functions kinetic and potential energy. These energies can be described as time and position dependent variables for a deformed object. The mathematical relationship of Hamilton's principle is defined as follows:

$$\delta \Pi = \int_T (\delta U + \delta W - \delta K) dt = 0. \quad (28)$$

In Eq. (28), $\delta \Pi$ is the first variation of total potential energy, which is equal to zero at the equilibrium point, δU is the first variation of strain energy density caused by internal loads,

δW is the first variation of work done due to external loads, and δK is the first variation of kinetic energy of the system.

In order to obtain the variation of strain energy of the system, the integral over the volume of the strain energy for the elastic body is considered as:

$$\delta U = \iiint_V \sigma_{ij} \delta \epsilon_{ij} dV, \quad i, j = rr, \theta\theta, rz. \quad (29)$$

By expanding the Eq. (29) and placing the strain components and applying the δ operator, we will have:

$$\delta U = \iiint_V \left[\sigma_{rr} \left(\delta \frac{\partial u_0}{\partial r} + \frac{\partial w_0}{\partial r} \delta \frac{\partial w_0}{\partial r} + z \frac{\partial \varphi_r}{\partial r} + f(z) \frac{\partial \psi_r}{\partial r} \right) + \sigma_{\theta\theta} \left(\frac{u_0}{r} + z \delta \frac{\varphi_r}{r} + \frac{f(z)}{r} \delta \psi_r \right) + \sigma_{rz} \left(\delta \frac{\partial w_0}{\partial r} + \delta \varphi_r + \delta f(z) \right) \right] dr d\theta dz. \quad (30)$$

Finally, by integrating over the thickness and replacing the results of stress and moment and also integrating from all the terms of the concluded sentence using the law of integration by parts and by replacing the results in Eq. (30) we have:

$$\delta U = - \iint_A \left\{ \begin{aligned} & \left[N_{rr} + r \frac{\partial N_r}{\partial r} - N_{\theta\theta} \right] \delta u_0 + \\ & \left[r \frac{\partial M_{rr}}{\partial r} + M_{rr} - M_{\theta\theta} - r Q_{rz} \right] \delta \varphi_r + \\ & \left[r \frac{\partial R_{rr}}{\partial r} + R_{rr} - R_{\theta\theta} - r R_{rz} \right] \delta \psi_r + \\ & \left[\frac{\partial}{\partial r} \left(r N_{rr} \frac{\partial w_0}{\partial r} + r \frac{\partial Q_{rz}}{\partial r} + Q_{rz} \right) \right] \delta w_0 \end{aligned} \right\} dr d\theta + \oint_{\Gamma} \left\{ \begin{aligned} & \left[r N_{rr} d\theta \right] \delta u_0 + \\ & \left[r M_{rr} d\theta \right] \delta \varphi_r + \\ & \left[r R_{rr} d\theta \right] \delta \psi_r + \\ & \left[r N_{rr} \frac{\partial w_0}{\partial r} + r Q_{rz} \right] \delta w_0 \end{aligned} \right\} d\Gamma. \quad (31)$$

The variation of kinetic energy of the system can be expressed as:

$$\delta K = \iiint_V \rho(z) \left((\dot{u}_0 + z \dot{\varphi}_r + f(z) \dot{\psi}_r) \delta (\dot{u}_0 + z \dot{\varphi}_r + f(z) \dot{\psi}_r) + \dot{w}_0 \delta (\dot{w}_0) \right) dV. \quad (32)$$

According to the Eq. (32) we have

$$\int_0^T \delta K dt = \int_0^T \left\{ \iiint_V \rho(z) \left((\dot{u}_0 - z \dot{\varphi}_r - f(z) \dot{\psi}_r) \delta (\dot{u}_0 - z \dot{\varphi}_r - f(z) \dot{\psi}_r) + \dot{w}_0 \delta (\dot{w}_0) \right) dV \right\} dt. \quad (33)$$

By introducing the mass inertia term as:

$$I_k = \int_{-h/2}^{h/2} \rho(z) (1, z, z^2, f(z), zf(z), f^2(z)) dz, k = 0, 1, 2, 3, 4, 5, \quad (34)$$

Eq. (34) can be rewritten as:

$$\int_0^T \delta K dt = \int_0^T \int_A \left[I_0 \dot{u}_0 \delta \dot{u}_0 - I_1 \dot{u}_0 \delta \dot{\varphi}_r - I_3 \dot{u}_0 \delta \dot{\psi}_r + I_1 \dot{\varphi}_r \delta \dot{u}_0 + I_2 \dot{\varphi}_r \delta \dot{\varphi}_r + I_4 \dot{\varphi}_r \delta \dot{\psi}_r - I_3 \dot{\psi}_r \delta \dot{u}_0 + I_4 \dot{\psi}_r \delta \dot{\varphi}_r + I_5 \dot{\varphi}_r \delta \dot{\varphi}_r + I_6 \dot{w}_0 \delta \dot{w}_0 \right] dA dt. \quad (35)$$

Also, the variation of work due to the external loads can be expressed as:

$$\delta W = \iiint_V (F(t)) \delta w dV. \quad (36)$$

expanding Eq. (28) results in:

$$\delta \Pi = \int_0^T \delta U dt + \int_0^T \delta W dt - \int_0^T \delta K dt = 0. \quad (37)$$

Inserting the Eqs. (31), (35) and (36) into (37), one obtains the following equations of motion for the entire system:

$$\frac{\partial N_{rr}}{\partial r} + \left(\frac{N_{rr} - N_{\theta\theta}}{r} \right) = \left(I_0 \frac{\partial^2 u}{\partial t^2} + I_1 \frac{\partial^2 \varphi_r}{\partial t^2} + I_3 \frac{\partial^2 \psi_r}{\partial t^2} \right), \quad (38)$$

$$\frac{\partial M_{rr}}{\partial r} - Q_{rz} + \left(\frac{M_{rr} - M_{\theta\theta}}{r} \right) = \left(I_1 \frac{\partial^2 u_0}{\partial t^2} + I_2 \frac{\partial^2 \varphi_r}{\partial t^2} + I_4 \frac{\partial^2 \psi_r}{\partial t^2} \right), \quad (39)$$

$$\frac{\partial R_{rr}}{\partial r} - R_{rz} + \left(\frac{R_{rr} - R_{\theta\theta}}{r} \right) = \left(I_3 \frac{\partial^2 u_0}{\partial t^2} + I_4 \frac{\partial^2 \varphi_r}{\partial t^2} + I_5 \frac{\partial^2 \psi_r}{\partial t^2} \right), \quad (40)$$

$$\frac{\partial Q_{rz}}{\partial r} + \frac{Q_{rz}}{r} + \left(N_{rr} \frac{\partial^2 w_0}{\partial r^2} + \frac{N_{\theta\theta}}{r} \frac{\partial w_0}{\partial r} \right) + F(t) = \left(I_0 \frac{\partial^2 w_0}{\partial t^2} \right). \quad (41)$$

Using Eqs. (20)–(27), and following some straight forward mathematical manipulations, the equations of motion in terms of displacement field can be written as follows:

$$\begin{aligned} & A_{11} \left(\frac{\partial^2 u_0}{\partial r^2} + \left(\frac{1}{2} \right) \frac{\partial}{\partial r} \left(\frac{\partial w_0}{\partial r} \right)^2 \right) + A_{12} \left(\frac{\partial}{\partial r} \left(\frac{u_0}{r} \right) \right) \\ & + B_{11} \frac{\partial^2 \varphi_r}{\partial r^2} + B_{12} \left(\frac{\partial}{\partial r} \left(\frac{\varphi_r}{r} \right) \right) \\ & + C_{11} \left(\frac{\partial^2 \psi_r}{\partial r^2} \right) + C_{12} \left(\frac{\partial}{\partial r} \left(\frac{\psi_r}{r} \right) \right) \\ & \left\{ \left(\frac{\partial u_0}{\partial r} + \left(\frac{1}{2} \right) \left(\frac{\partial w_0}{\partial r} \right)^2 \right) (A_{11} - A_{12}) + \right. \\ & \left. + \frac{1}{r} \left\{ \frac{u_0}{r} (A_{12} - A_{22}) + \frac{\partial \varphi_r}{\partial r} (B_{11} - B_{12}) + \right. \right. \\ & \left. \left. \frac{\varphi_r}{r} (B_{12} - B_{22}) + \frac{\partial \psi_r}{\partial r} (C_{11} - C_{12}) + \right. \right. \\ & \left. \left. \frac{\psi_r}{r} (C_{12} - C_{22}) \right\} \right. \\ & - A_1 (I_0 u_{0j+1} + I_1 \varphi_{rj+1} + I_3 \psi_{rj+1}) = \\ & - I_0 (A_1 u_{0j} + A_2 \dot{u}_{0j} + A_3 \ddot{u}_{0j}) + I_1 (A_1 \varphi_{rj} + A_2 \dot{\varphi}_{rj} + A_3 \ddot{\varphi}_{rj}) \\ & + I_3 (A_1 \psi_{rj} + A_2 \dot{\psi}_{rj} + A_3 \ddot{\psi}_{rj}), \end{aligned} \quad (42)$$

$$\begin{aligned} & B_{11} \left(\frac{\partial^2 u_0}{\partial r^2} + \left(\frac{1}{2} \right) \frac{\partial}{\partial r} \left(\frac{\partial w_0}{\partial r} \right)^2 \right) + B_{12} \left(\frac{\partial}{\partial r} \left(\frac{u_0}{r} \right) \right) \\ & + F_{11} \frac{\partial^2 \varphi_r}{\partial r^2} + F_{12} \left(\frac{\partial}{\partial r} \left(\frac{\varphi_r}{r} \right) \right) + \\ & G_{11} \left(\frac{\partial^2 \psi_r}{\partial r^2} \right) + G_{12} \left(\frac{\partial}{\partial r} \left(\frac{\psi_r}{r} \right) \right) + \\ & \left\{ \left(\frac{\partial u_0}{\partial r} + \left(\frac{1}{2} \right) \left(\frac{\partial w_0}{\partial r} \right)^2 \right) (B_{11} - B_{12}) + \right. \\ & \left. \frac{1}{r} \left\{ \frac{u_0}{r} (B_{12} - B_{22}) + \frac{\partial \varphi_r}{\partial r} (F_{11} - F_{12}) + \right. \right. \\ & \left. \left. \frac{\varphi_r}{r} (F_{12} - F_{22}) + \frac{\partial \psi_r}{\partial r} (G_{11} - G_{12}) + \right. \right. \\ & \left. \left. \frac{\psi_r}{r} (G_{12} - G_{22}) \right\} \right. \\ & D \left(\varphi_r + \frac{\partial w_0}{\partial r} \right) - E(\psi_r) - A_1 (I_1 u_{0j+1} + I_2 \varphi_{rj+1} + I_4 \psi_{rj+1}) = \\ & - I_1 (A_1 u_{0j} + A_2 \dot{u}_{0j} + A_3 \ddot{u}_{0j}) + I_2 (A_1 \varphi_{rj} + A_2 \dot{\varphi}_{rj} + A_3 \ddot{\varphi}_{rj}) \\ & + I_4 (A_1 \psi_{rj} + A_2 \dot{\psi}_{rj} + A_3 \ddot{\psi}_{rj}), \end{aligned} \quad (43)$$

Table 2 The initial and boundary conditions for two simple and clamped supports on annular plates

	aaa	aaa	aaa	aaa	aaa	aaa
Initial conditions	0	0	0	0	–	–
Simply supports	0	0	–	–	0	0
Boundary conditions						
Clamped	0	0	0	0	–	–

$$\begin{aligned}
& C_{11} \left(\frac{\partial^2 u_0}{\partial r^2} + \left(\frac{1}{2} \right) \frac{\partial}{\partial r} \left(\frac{\partial w_0}{\partial r} \right)^2 \right) + C_{12} \left(\frac{\partial}{\partial r} \left(\frac{u_0}{r} \right) \right) \\
& + G_{11} \frac{\partial^2 \varphi_r}{\partial r^2} + G_{12} \left(\frac{\partial}{\partial r} \left(\frac{\varphi_r}{r} \right) \right) + \\
& H_{11} \left(\frac{\partial^2 \psi_r}{\partial r^2} \right) + H_{12} \left(\frac{\partial}{\partial r} \left(\frac{\psi_r}{r} \right) \right) + \\
& \left\{ \left(\frac{\partial u_0}{\partial r} + \left(\frac{1}{2} \right) \left(\frac{\partial w_0}{\partial r} \right)^2 \right) (C_{11} - C_{12}) + \right. \\
& \left. \frac{1}{r} \left(\frac{u_0}{r} (C_{12} - C_{22}) + \frac{\partial \varphi_r}{\partial r} (G_{11} - G_{12}) + \right. \right. \\
& \left. \left. \frac{\varphi_r}{r} (G_{12} - G_{22}) + \frac{\partial \psi_r}{\partial r} (H_{11} - H_{12}) + \right. \right. \\
& \left. \left. \frac{\psi_r}{r} (H_{12} - H_{22}) \right) \right\} - \\
& e \left(\varphi_r + \frac{\partial w_0}{\partial r} \right) - I(\psi_r) - A_1 (I_3 u_{0j+1} + I_2 \varphi_{rj+1} + I_5 \psi_{rj+1}) = \\
& -I_3 (A_1 u_{0j} + A_2 \dot{u}_{0j} + A_3 \ddot{u}_{0j}) + I_2 (A_1 \varphi_{rj} + A_2 \dot{\varphi}_{rj} + A_3 \ddot{\varphi}_{rj}) \\
& + I_5 (A_1 \psi_{rj} + A_2 \dot{\psi}_{rj} + A_3 \ddot{\psi}_{rj}), \quad (44)
\end{aligned}$$

$$\begin{aligned}
& \left[A_{11} \left(\frac{\partial u_0}{\partial r} + \left(\frac{1}{2} \right) \left(\frac{\partial w_0}{\partial r} \right)^2 \right) \right] \frac{\partial^2 w_0}{\partial r^2} + \\
& A_{12} \left(\frac{u_0}{r} \right) + B_{11} \left(\frac{\partial \varphi_r}{\partial r} \right) + B_{12} \left(\frac{\varphi_r}{r} \right) \\
& + C_{11} \left(\frac{\partial \psi_r}{\partial r} \right) + C_{12} \left(\frac{\psi_r}{r} \right) + \\
& \left[A_{12} \left(\frac{\partial u_0}{\partial r} + \left(\frac{1}{2} \right) \left(\frac{\partial w_0}{\partial r} \right)^2 \right) \right] \frac{1}{r} \frac{\partial w_0}{\partial r} + \\
& A_{22} \left(\frac{u_0}{r} \right) + B_{12} \left(\frac{\partial \varphi_r}{\partial r} \right) + B_{22} \left(\frac{\varphi_r}{r} \right) \\
& + C_{12} \left(\frac{\partial \psi_r}{\partial r} \right) + C_{22} \left(\frac{\psi_r}{r} \right) + \\
& D \left(\left(\frac{\partial \varphi_r}{\partial r} + \frac{\partial^2 w_0}{\partial r^2} \right) + \left(\frac{\varphi_r}{r} + \frac{\partial w_0}{r \partial r} \right) \right) \\
& + E \left(\frac{\partial \psi_r}{\partial r} + \frac{\psi_r}{r} \right) - A_1 (I_0 w_{0j+1}) = \\
& -I_0 (A_1 w_{0j} + A_2 \dot{w}_{0j} + A_3 \ddot{w}_{0j}) - F(t). \quad (45)
\end{aligned}$$

To complete the formulations, the equilibrium equations are accompanied by a set of boundary conditions as well as initial conditions. In Table 2, the initial and boundary conditions for two simple and clamped supports on annular plates are investigated.

5 Solving technique

In order to discretize the time-dependent equations of motion of an FG annular porous plate, the implicit Newmark method is used in this paper, and to solve the partial differential equations of motion, the viscous dynamic relaxation method (V-DR) technique is employed.

5.1 Newmark integration

The Newmark integration method is an implicit numerical method for the discretization of time derivatives performed using the reduced Taylor series. This method has been used in many dynamic analyses of structures because of its simplicity. The purpose of using this method in this study is to discretize partial differential equations in the time domain and solve them separately. In this method, the solution of the derivatives in the time interval is approximated up to the second order using the reduced Taylor series.

The main goal of the Newmark approach is to discretize the time-varying equations using a reduced Taylor series by determining the accelerations and velocities in real forms to the next time step (t_{j+1}) as [35]:

$$\dot{X}_{j+1} = \left\{ \left(\frac{\gamma}{\beta \Delta t_j} \right) (X_{j+1} - X_j) - \left(\frac{\gamma}{\beta} - 1 \right) \dot{X}_j - \left(\frac{\gamma}{2\beta} - 1 \right) \Delta t_j \ddot{X}_j \right\}, \quad (46)$$

$$\ddot{X}_{j+1} = \left\{ \left(\frac{1}{\beta (\Delta t_j)^2} \right) (X_{j+1} - X_j) - \left(\frac{1}{\beta \Delta t_j} \right) \dot{X}_j - \left(\frac{1}{2\beta} - 1 \right) \ddot{X}_j \right\}. \quad (47)$$

In the above equations, j represents the time step. Also, β and γ are the Newmark constants that should be chosen to reduce the error in the calculations so that the responses of the next step are calculated correctly. If these constant parameters can be selected to minimize the error, stable solutions can be provided. The Newmark method is stable enough under certain conditions that it is stable for all values of $2\beta > \gamma > 0.5$ or the values of $\beta \leq \gamma$ and $\gamma \geq 0.5$ are conditionally stable. In this work, the values of β and γ for the average acceleration are assumed to be 0.25 and 0.5, respectively. Additionally, Δt is the time interval, the difference between current and previous real-time displacement, velocity, and acceleration, respectively. Moreover, X represents the displacements and rotations ($X = u_0, w_0, \varphi_r$ and ψ_r) at $t_{(j+1)}$ and t_j .

Substituting Eqs. (46) and (47) into (38), (39), (40) and (41) results in the dynamic form of the equations of motion as follows:

$$\frac{\partial N_{rr}}{\partial r} + \left(\frac{N_{rr} - N_{\theta\theta}}{r} \right) - A_1 (I_0 u_{0j+1} + I_1 \varphi_{rj+1} + I_3 \psi_{rj+1}) = -I_0 (A_1 u_{0j} + A_2 \dot{u}_{0j} + A_3 \ddot{u}_{0j}) + I_1 (A_1 \varphi_{rj} + A_2 \dot{\varphi}_{rj} + A_3 \ddot{\varphi}_{rj}) - I_3 (A_1 \psi_{rj} + A_2 \dot{\psi}_{rj} + A_3 \ddot{\psi}_{rj}), \quad (48)$$

$$\frac{\partial M_{rr}}{\partial r} - Q_{rz} + \left(\frac{M_{rr} - M_{\theta\theta}}{r} \right) - A_1 (I_1 u_{0j+1} + I_2 \varphi_{rj+1} + I_4 \psi_{rj+1}) + I_1 (A_1 u_{0j} + A_2 \dot{u}_{0j} + A_3 \ddot{u}_{0j}) + I_2 (A_1 \varphi_{rj} + A_2 \dot{\varphi}_{rj} + A_3 \ddot{\varphi}_{rj}) - I_4 (A_1 \psi_{rj} + A_2 \dot{\psi}_{rj} + A_3 \ddot{\psi}_{rj}), \quad (49)$$

$$\frac{\partial R_{rr}}{\partial r} - R_{rz} + \left(\frac{R_{rr} - R_{\theta\theta}}{r} \right) - A_1 (I_3 u_{0j+1} + I_2 \varphi_{rj+1} + I_5 \psi_{rj+1}) = -I_3 (A_1 u_{0j} + A_2 \dot{u}_{0j} + A_3 \ddot{u}_{0j}) + I_2 (A_1 \varphi_{rj} + A_2 \dot{\varphi}_{rj} + A_3 \ddot{\varphi}_{rj}) + I_5 (A_1 \psi_{rj} + A_2 \dot{\psi}_{rj} + A_3 \ddot{\psi}_{rj}), \quad (50)$$

$$\frac{\partial Q_{rz}}{\partial r} + \frac{Q_{rz}}{r} + \left(N_{rr} \frac{\partial^2 w_0}{\partial r^2} + \frac{N_{\theta\theta}}{r} \frac{\partial w_0}{\partial r} \right) - A_1 (I_0 w_{0j+1}) = -I_0 (A_1 w_{0j} + A_2 \dot{w}_{0j} + A_3 \ddot{w}_{0j}) - F(t), \quad (51)$$

where the Newmark coefficients are defined as:

$$A_1 = \frac{1}{\beta(\Delta t_j)^2}, A_2 = \frac{1}{\beta \Delta t_j}, A_3 = \frac{1}{2\beta} - 1. \quad (52)$$

For the sake of brevity, we may write the following shrunken term for the equations of motion:

$$[\bar{K}_{j+1}] X_{j+1} = \bar{P}_{j+1}, \quad (53)$$

in which the terms $\bar{K}_{j+1}$ and $\bar{P}_{j+1}$ are the equivalent stiffness matrix and load vector in the n^{th} iteration of the V-DR procedure.

5.2 Viscous dynamic relaxation

On the basis of the dynamic relaxation method, the governing equations of motion (53) are transformed from a static space to a fictitious dynamic space in order to obtain stable solutions by converting boundary value problems to initial value problems. The conversion is done by adding damping and inertia terms for the next time step to the equilibrium equations [36]:

$$[M]_{j+1}^n \{\ddot{X}\}_{j+1}^n + [C]_{j+1}^n \{\dot{X}\}_{j+1}^n + [K]_{j+1}^n \{X\}_{j+1}^n = \{P(t_{j+1})\}^n. \quad (54)$$

In Eq. (54), $[M]$ and $[C]$ are fictitious mass and damping matrices, respectively. Also, X , $\dot{X}$ and $\ddot{X}$ represent displacement, velocity, and acceleration vectors. The presence of damping in the viscous dynamic release method causes

the transient response to tend to zero after a period of time. It should be noted that this period of time depends on the level of damping of the system. So, in a dynamic system, static loading is introduced, a stable state of the problem is obtained with the passage of time, and with the correct selection of this amount of damping, it is also possible to reach a faster convergence. Now, using the finite difference method, the velocity and acceleration vectors are achieved as [36]:

$$\{\ddot{X}\}^n = \frac{\{\dot{X}\}^{n+0.5} - \{\dot{X}\}^{n-0.5}}{\tau^n}, \quad (55)$$

$$\{\dot{X}\}^{n-0.5} = \frac{\{X\}^n - \{X\}^{n-1}}{\tau^n}. \quad (56)$$

The criterion of convergence in the dynamic relaxation method is the correct choice and guess of the mass and damping matrices, so for this purpose, using the Gershgorin theory, this mass matrix can be expressed as follows [36]:

$$m_{ii}^d \geq \frac{1}{4} [\tau^n]^2 \sum_{l=1}^N |k_{il}|. \quad (57)$$

For the damping matrix, we also have [36]:

$$c_i^n = 2 \left[\frac{(X_i^n)^T \{\bar{P}_{j+1}\}_i^n}{(X_i^n)^T m_{ii}^d X_i^n} \right]^{0.5}. \quad (58)$$

In Eqs. (57) and (58), d denote the number of degrees of freedom of the structure, which in this research is 4 (u_0 , w_0 , φ_r and w_0). i in mass and damping relations also represents the desired node in each iteration of the dynamic relaxation method. Also, the stiffness matrix k_{il} of the whole system is calculated from the following Eq. [36]:

$$k = \frac{\partial \bar{P}_{j+1}}{\partial X}. \quad (59)$$

c_i^n is the critical damping coefficient at i^{th} node, τ^n indicating fictitious incremental time considered unity in this work. Now the velocity for the next time step can be defined as follows [37]:

$$\{\dot{X}\}^{n+0.5} = \frac{(2m_{ii}^d - \tau^n c_i^n)}{(2m_{ii}^d + \tau^n c_i^n)} \{\dot{X}\}^{n+0.5} + \frac{2\tau^n}{(2m_{ii}^d + \tau^n c_i^n)} \{R\}^n, \quad (60)$$

in which, $\{R\}^n$ is the residual forces or the values on the left side in the equilibrium relations of the system. The residual forces and the total kinetic energy of the system can be defined as follows [38]:

$$\{R\}^n = \{\bar{P}_{j+1}\}^n - [\bar{K}_{j+1}]\{X\}^n, \quad (61)$$

$$KE^{n+1} = \sum_{i=1}^N \{\dot{X}_i^{n+0.5}\}^2. \quad (62)$$

By using the following equation, displacement and rotations can be calculated at the end of each time increment [36]:

$$X^{n+1} = X^n + \tau^{n+1} \dot{X}^{n+0.5}. \quad (63)$$

The V-DR process continues with iterative steps to meet the desired convergence criteria that is $KE^{n+1} \leq 10^{-12}$ and $(R)^n \leq 10^{-6}$. The following steps explain the solving procedure loop of V-DR method with the Newmark technique combined:

1. Specify total number of time steps (N_{tot}) and initial conditions (all variables set to zero), afterwards $\Rightarrow i = 1$,
2. Set previously converged displacements,
3. Form artificial mass and damping matrices,
4. Indicate fictitious velocity vector using Eq. (56),
5. Calculate displacements using Eq. (63) and apply boundary conditions,
6. Check whether the convergency criteria are satisfied i.e., $KE^{n+1} \leq 10^{-12}$ and $(R)^n \leq 10^{-6}$,
7. Are the criterion factors achieved?
 $\left\{ \begin{array}{l} \text{No} \rightarrow \text{go to 3} \\ \text{Yes} \rightarrow \text{Continue} \end{array} \right\},$
8. Calculate $\ddot{X}_{j+1}$ and $\dot{X}_{j+1}$ using Eqs. (46) and (47) and set $i = i + 1$,
9. Is $i > N_{tot}$? $\left\{ \begin{array}{l} \text{No} \rightarrow \text{Go to (b)} \\ \text{Yes} \rightarrow \text{Print out the results} \end{array} \right\}$
10. stop.

6 Discussions and numerical results

6.1 Comparison study

To illustrate the accuracy and precision of the presented method, Fig. 3 shows the comparison of the results of ref. [39] for an annular FG Mindlin plate theory under impulsive loading and simply-simply supported conditions with the present study modeling. Based on a Mori–Tanaka scheme, the Poisson's ratio varies through the thickness and the below terms are taken as assumptions of the problem:

$$E_m = 70 \text{ GPa}, \nu_m = 0.3, \rho_m = 2702 \frac{\text{kg}}{\text{m}^3}, \quad (64)$$

$$E_c = 427 \text{ GPa}, \nu_c = 0.17, \rho_c = 3100 \frac{\text{kg}}{\text{m}^3}, \quad (65)$$

$$\frac{r_o}{h} = 6, \frac{r_i}{h} = 3, R = \frac{(r_o + r_i)}{2}, \quad (66)$$

$$T = \frac{t}{\Delta t}, \quad (67)$$

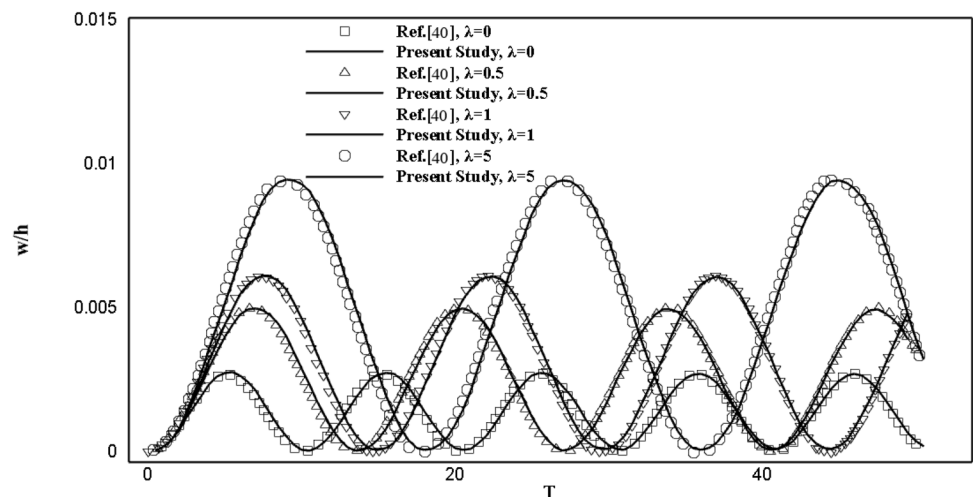
$$\Delta t = 0.01 \text{ s}, \quad (68)$$

$$F(t) = q_0 = 0.42 \text{ GPa}, (t < \infty), \quad (69)$$

in which R is the normalized radius point, w/h is representing nondimensional deflection and $T = t/\Delta t$ is nondimensional time.

The results are a good indicator of great consistency between the present study and the Ref. [39]

Fig. 3 Comparison study for dynamic response of simply supported FG annular plate having different power law indices under impulsive loading. $\frac{r_o}{h} = 6, \frac{r_i}{h} = 3, F(t) = q_0 = 0.42 \text{ GPa} (t < \infty)$



6.2 Parametric study

This section is devoted to studying the effect of some key factors such as boundary conditions, dynamic loadings, power-law indices, porosity patterns, and volume fractions on the dynamic behavior of FGP annular plates. In order to study the annular plates, the plates with two aspect ratios of $h/r_0 = 0.3, 0.15$, with and outer and inner radius of $r_0 = 0.1$ m, $r_i = 0.02$ m are assumed as the geometric parameters of the problem in this section. Moreover, the plates are under three types of dynamic loadings namely impulsive, impact, and harmonic as relations (70), (71), and (72).

$$\text{Impact loading: } F(t) = \begin{cases} q_0 t, & t \leq t_{\text{tot}} \\ 0, & t > t_{\text{tot}} \end{cases}, \quad (70)$$

$$\text{Impulsive loading: } F(t) = q_0, \quad t < \infty, \quad (71)$$

$$\text{Harmonic loading: } F(t) = q_0 \sin(20t), \quad t < \infty. \quad (72)$$

For the entire parametric study in this section, the following assumptions are made for dimensionless dynamic and radial stress analysis at a normalized R point:

$$t_{\text{tot}} = 0.3 \text{ s}, T = \frac{t}{\Delta t}, R = \frac{(r_o + r_i)}{2}, q_0 = 15 \text{ Kpa}, z = 0. \quad (73)$$

In which $z = 0$ is the midpoint plate from distance R

Figure 4 shows the effect of boundary supports on the dimensionless deflection of the center of the annular porous plate subjected to an impulse load with a magnitude of 15 MPa. With a uniform porosity pattern and simple or clamped edge supports, the plate shows up to 6% better resistance to vibration when clamped. In other words, without porosity, when the power law index was changed from 0 to 5, an increase of 80% was observed for the clamped support and 84.21% for the simple support as is shown in Fig. 4b. Figure 4c. On the other hand, when the porosity was changed from 0 to 5 and the power-law index was 0.5, Fig. 4d showed an increase of 60.8% and 66.66% in each vibration peak. The type of support (simple or clamped) and the presence of uniform porosity strongly affect the plate's vibration resistance. Clamped supports offer significantly better vibration resistance than simple supports. Increasing the power law indices enhances resistance in both cases when there is no porosity. Also, the growing porosity while adjusting the power-law index boosts vibration resistance, as shown by changes in deflection and vibration peaks.

Figure 5 discusses the influence of various thickness-to-radius ratios on the dynamic deflection and dimensionless radial stress of the FGP plate under impulse loading

and simple and clamped edge supports. The porosity volume fraction and the power law index are assumed to be 0.5. Figure 5a and b show that with increasing plate thickness, the mean deflection of the plates decreases by 159.49% in clamped and 166.99% in simple support. Figure 5c and d show the effect of thickness-to-radius on the dimensionless radial stress at $z = 0$. The results show that the stress decreases with decreasing thickness up to 9.26% for clamped, while it is consistent, and 11.54% for simple supports, but this is only true for the first peak and can vary over the next peaks and even become higher than for the thicker plates. The aim is to discuss how the thickness-to-radius ratio affects the behavior of a FGP plate subjected to impulse loading with different support conditions. Thicker plates experience reduced deflection, with a more pronounced effect in clamped support than in simple support. Additionally, thinner plates exhibit lower radial stress at specific conditions, up to a certain point, after which the stress behavior varies in subsequent peaks. Thicker plates, in contrast, may show different stress patterns, potentially even experiencing higher stress in later peaks.

Figure 6 studies the effect of different porosity patterns on dimensionless dynamic deflection and radial stress of the plate at $z = 0$ under harmonic loading with a magnitude of 15 MPa. Figure 6a and b show that in both boundary conditions, with the same amount of pore volume fraction, porosity pattern X acts best in a decrease of deflection compared to the other two porosity patterns. In clamped conditions, changing patterns X to uniform applies 7.27% and X to O, up to 29.3% of deflection reduction, and in simply supported plate changes of patterns X to uniform gives 8% and X to O of almost 15.38% deflection decrease was observed. Furthermore, Fig. 6c and d discuss the effect of each pattern on the nondimensional radial stress of the porous FG annular plate. It is found that with a change in patterns from X to U in clamped support, a 52.20% stress decrease and 111.59% from X to O were observed. Moreover, in simple support changes from pattern X to O and X to U show 56.45% and 118.61% of stress reduction, respectively. Figure 6 examines different porosity patterns' impact on plate's dynamic deflection and radial stress under harmonic loading. Switching to uniform (U) or pattern O from X significantly reduces deflection and radial stress in clamped and simple supports. In summary, altering the porosity pattern greatly reduces deflection and stress in the plate.

Figures 7 and 8 are dedicated to representing the effect of pore volume fraction in each pattern over nondimensional deflection and radial stress of the porous FG plate under impact loading. As it is evident in Fig. 7, with an increase in porosity, the deflection of the three dispersions greatens in relation to each other. It can be concluded

Fig. 4 Time history of FG porous annular plate under impulsive loading with $q_0 = 15$ Kpa, uniform porosity pattern with different ξ and α

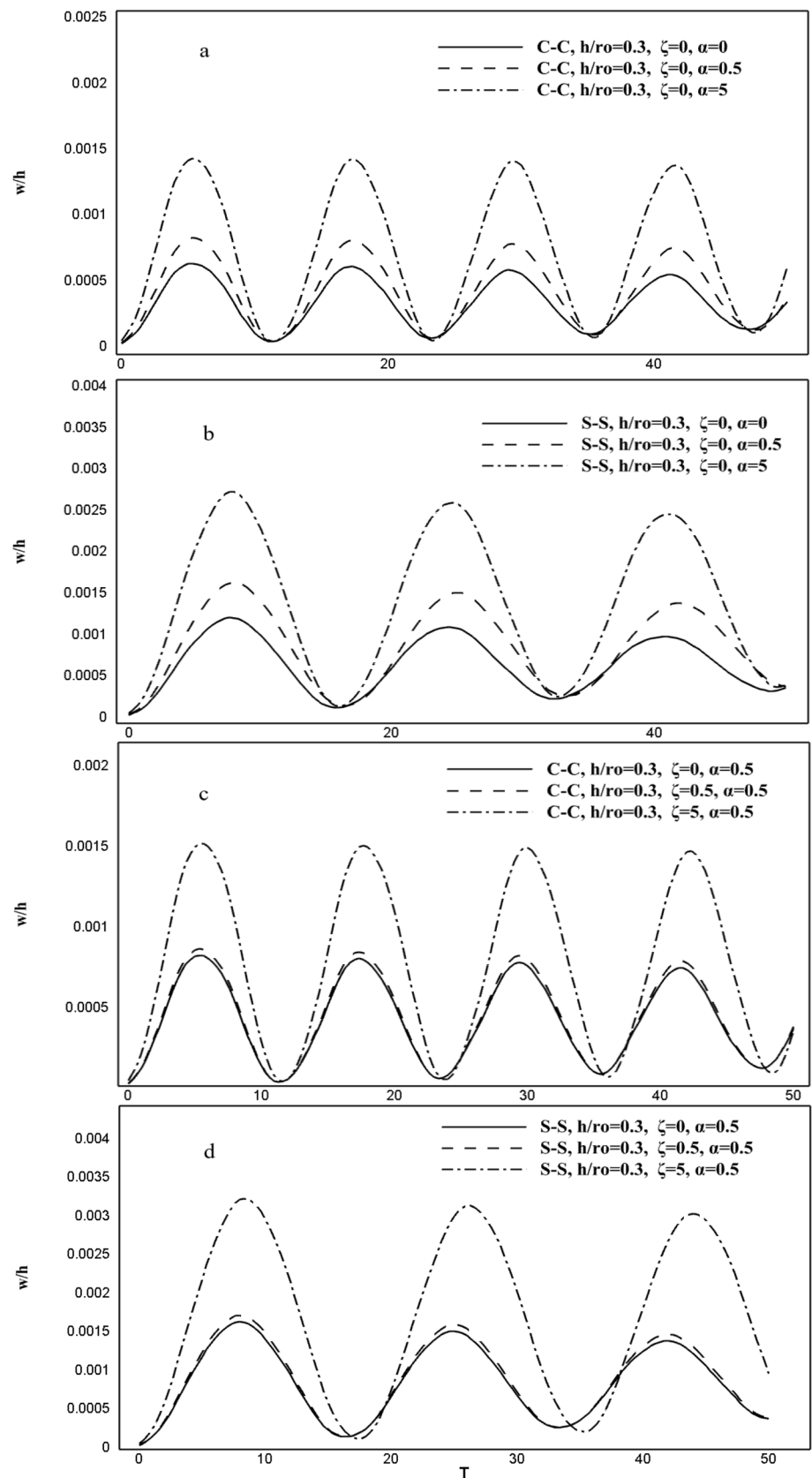


Fig.5 Nondimensional Radial Stress and dynamic deflection of FG porous annular plate under impulsive loading with $q_0 = 15$ Kpa, uniform porosity pattern

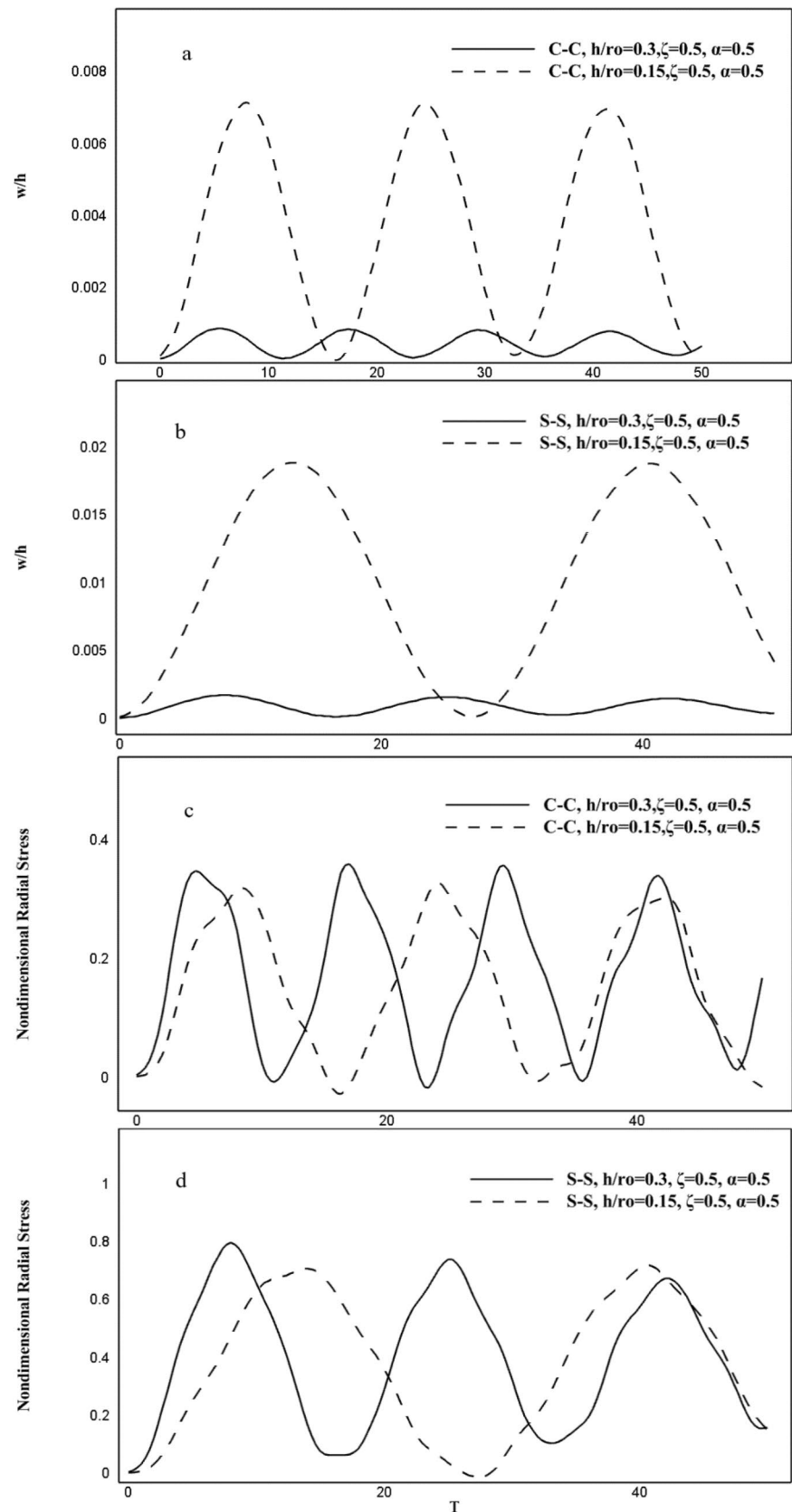


Fig.6 Effect of different porosity patterns on Non-dimensional Radial Stress and dynamic deflection of FG porous annular plate under harmonic loading with $q_0 = 15 \text{ Kpa}$, $h/r_0 = 0.3$

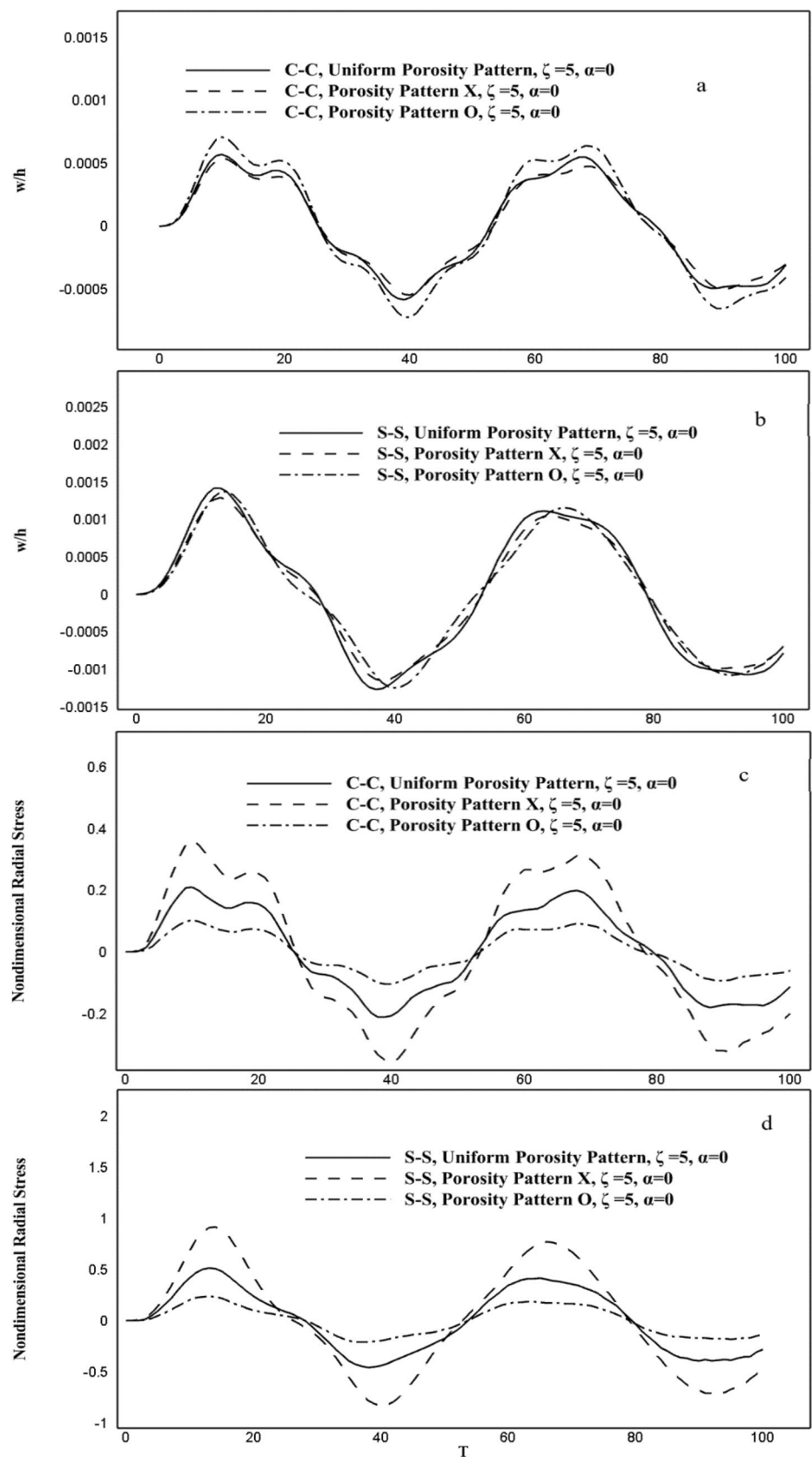
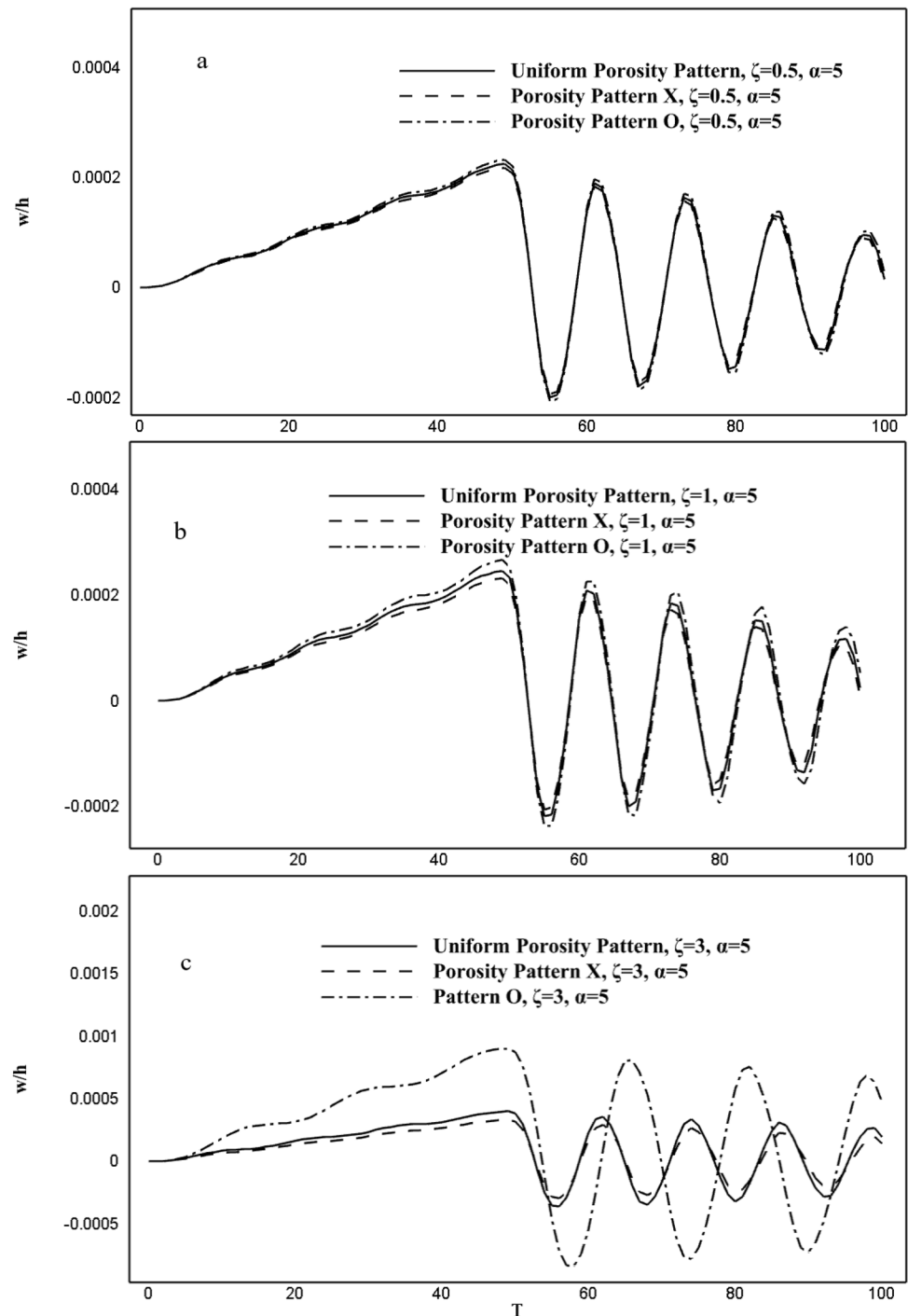


Fig. 7 Effect of different porosity patterns and different porosity volume fraction on Nondimensional dynamic deflection of FG porous annular plate under Impact loading and Clamped condition, $h/r_0 = 0.3$

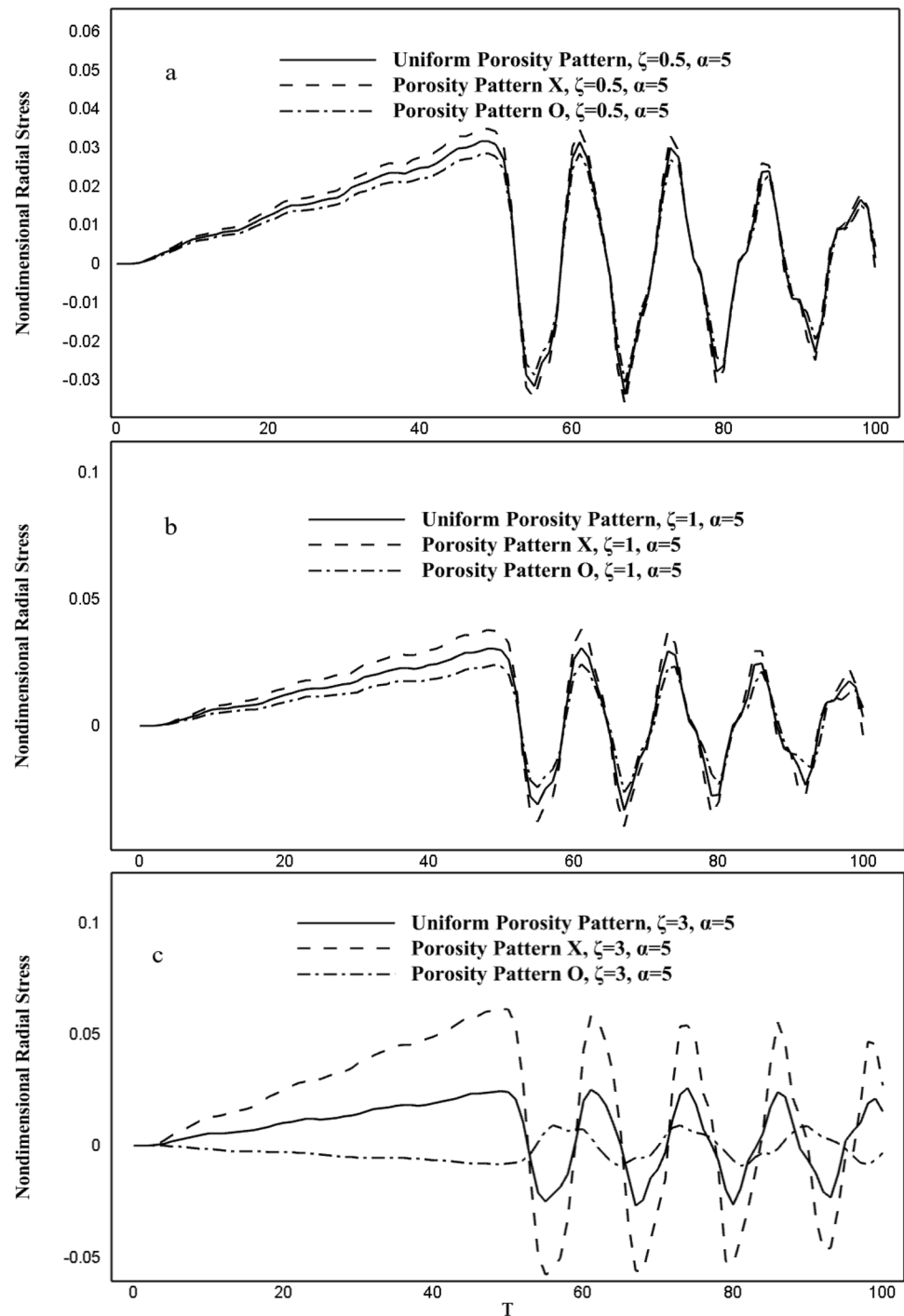


that the porosity dispersion X has the best resistance for deflection with an increase in the pore volume. On the other hand, porosity O shows almost 79.53% increase in deflection compared to pattern U and 95.99% compared with pattern X. But Fig. 8 shows the superiority of the pattern O over patterns X and U in other ways, that it was observed that, an increase of pore fraction from changes of pattern X to U gives 87.13% of stress reduction and up to 154% of lesser stress from pattern X to O. The outcome of

the Figs. 7 and 8 depict that increasing porosity amplifies deflection for all patterns. Pattern X shows superior resistance to deflection with rising pore volume, while pattern O is notably more vulnerable. Pattern O excels in reducing stress compared to patterns X and U, with substantial stress reduction when transitioning from X to O.

In order to conduct a comparison between linear and non-linear analyses of the PFG plates, the results for two different load magnitudes in Fig. 9 are presented.

Fig. 8 Effect of different porosity patterns and different porosity volume fraction on Nondimensional dynamic deflection of FG porous annular plate under Impact loading and Clamped condition, $h/r_0 = 0.3$



As depicted, there is a discernible difference of 4.77% between the linear and non-linear deflection with simply support conditions. This discrepancy highlights that as the load factor increases, the disparity between the linear and non-linear analyses becomes more pronounced. This divergence is primarily attributed to the fact that non-linear analysis considers various factors, such as large deformations, material nonlinearity, and alterations in boundary conditions, which can lead to a reduction in the structure's

stiffness and load-carrying capacity. In contrast, linear analysis operates under the assumption that the structure behaves in a linear and elastic manner, where displacements are directly proportional to the applied loads and material properties remain constant. However, it is important to note that linear analysis may underestimate the deflection of the structure when these assumptions do not accurately represent the real-world behavior of the system.

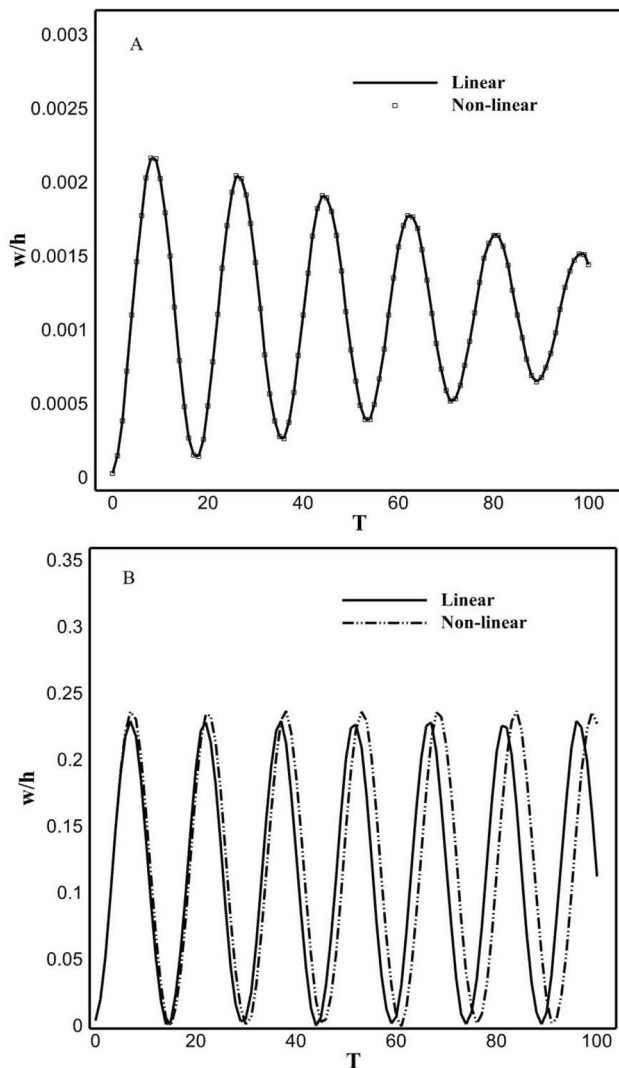


Fig. 9 Effect of different load magnitude on dynamic deflection of FG porous annular plate under harmonic loading with **A** $q_0 = 15$ Kpa and **B** $q_0 = 2000$ Kpa, $h/r_0 = 0.3$

To carry out analysis for clamped-simply and simply-clamped conditions plates, the results of a PFG plate under impulsive loading are presented in Fig. 10. The results depict that there is a difference of 28.24% between the two restrictions, saying that the S–C condition has more stiffness in a more extended area, so the bending of the plate will reduce.

In the end, it can be concluded that porosity pattern X shows the best reinforcing behavior for the plates and porosity pattern O acts as the best stress reducer in FGP material.

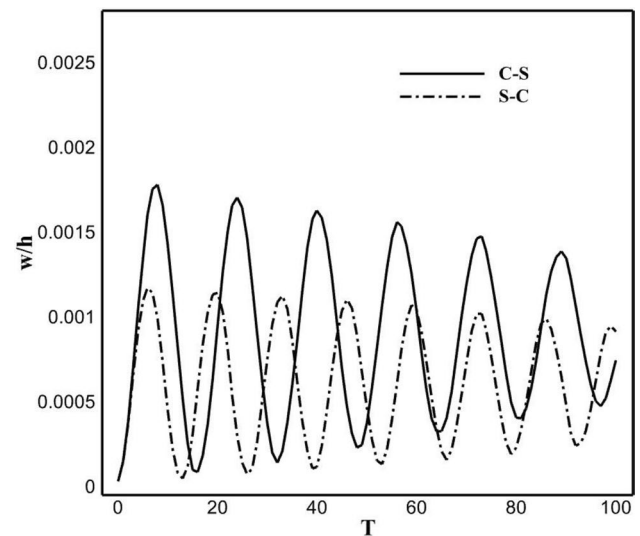


Fig. 10 Effect of different boundary conditions on dynamic deflection of FG porous annular plate under harmonic loading with $q_0 = 15$ Kpa, and $h/r_0 = 0.3$

7 Conclusion

This study undertook a comprehensive examination of the behavior of a porous plate under various loading conditions. The investigation focused on analyzing the effects of different factors, including boundary supports, thickness-to-radius ratios, porosity patterns, and porosity volume fraction, on the plate's dimensionless deflection and radial stress. The experiment involved a meticulous assessment of how each parameter impacts the mechanical properties of the porous plate. The results of this research provide valuable insights into the design and engineering of porous materials that can withstand different types of loads and stresses. Overall, this study sheds light on how to create stronger porous materials that are capable of enduring various types of stresses.

In mechanical systems, boundary supports play a crucial role in maintaining structural stability. Recent study has indicated that clamped edge supports are more effective than simple supports in increasing the system's resistance to vibrations, potentially resulting in an improvement of up to 6%. This is particularly significant in systems with higher power-law indices, where achieving vibration resistance can be challenging. As a result, it is recommended that implementing clamped edge supports be considered as a strategy to enhance the durability and stability of mechanical systems.

The impact of thickness-to-radius ratios on the structural behavior of a plate cannot be understated. As the thickness of the plate increases, the mean deflection experiences a significant decrease. Conversely, a reduction in

thickness leads to a decrease in radial stress, which is primarily responsible for the first stress peak. It is important to note that the first stress peak is the point at which the plate is most likely to fail. Thus, a thorough understanding of the relationship between thickness-to-radius ratios and plate behavior is vital in ensuring the safety and reliability of structures that rely on plates for support.

The present study aimed to investigate the porosity patterns of two distinct materials, pattern X and pattern O, with a view to evaluating their respective performances in reducing deflection and radial stress. The study results indicate that pattern X demonstrates superior performance in minimizing deflection, exhibiting reductions of up to 29.3% and 15.38% in clamped and simple supports, respectively. Conversely, pattern O outperforms pattern X in reducing radial stress, exhibiting up to 154% lower stress than pattern X. These findings suggest that the choice of porosity pattern should be made based on the specific performance requirements of the given material.

The porosity volume fraction of a material, which refers to the amount of empty space within it, plays a crucial role in its deflection under load. Higher porosity typically leads to greater deflection, although the choice of pattern used to structure the material can significantly impact its ability to resist deflection. Recent studies have indicated that pattern X provides optimal resistance to deflection, while pattern O exhibits superior stress reduction capabilities that make it an ideal choice for materials subjected to high levels of stress.

The results of the study show that the porosity pattern X, which was observed to have a higher level of interconnected pores, led to an increased resistance to deflection. On the other hand, pattern O, which had a more uniform pore size distribution, was found to effectively reduce stress in FG porous materials. These insights provide valuable information that can be used to optimize plate design and enhance performance.

Author contributions A wrote the main manuscript text and B provided the Matlab code. All authors reviewed the manuscript.

Data availability The authors confirm that the data supporting the findings of this study are available within the article.

Declarations

Conflict of interest The authors did not receive support from any organization for the submitted work. The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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